

Hunjae Lee

PhD Student in Machine Learning · Southern Methodist University

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EDUCATION

- | | |
|--|----------------|
| Ph.D. in Machine Learning | 2024 – Present |
| <i>Lyle School of Engineering, Southern Methodist University, Dallas, TX</i> | |
| <ul style="list-style-type: none">Fully funded fellowshipAdvised by Dr. Corey Clark | |
| M.S. in Computer Science | 2022 – 2024 |
| <i>Lyle School of Engineering, Southern Methodist University, Dallas, TX</i> | |
| <ul style="list-style-type: none">Fully funded fellowshipAdvised by Dr. Corey Clark | |
| B.S. in Computer Science | 2018 – 2022 |
| <i>Lyle School of Engineering, Southern Methodist University, Dallas, TX</i> | |
| <ul style="list-style-type: none">Lyle Distinguished Scholar | |

PUBLICATIONS

- **Hunjae Lee**, Corey Clark. **On the Runway Cascade of Transformers for Language Modeling**. *Under Review* 2026. [arXiv:2601.14522](https://arxiv.org/abs/2601.14522)
- **Hunjae Lee**, Corey Clark. **Dynamic Relational Priming Improves Transformer in Multivariate Time Series**. *International Conference on Machine Learning (ICML)*, 2026. [arXiv:2509.12196](https://arxiv.org/abs/2509.12196)
- Kassi Nzalasse, **Hunjae Lee**, Corey Clark. **KADI: Knowledge & Ability Deployment Infrastructure**. *Under Review* 2026.
- **Hunjae Lee**, Corey Clark. **Transformer Modeling for Both Scalability and Performance in Multivariate Time Series**. *Under Review* 2025. [arXiv:2509.19471](https://arxiv.org/abs/2509.19471)
- **Hunjae Lee**, Corey Clark. **Blind Evaluation Framework for Fully Homomorphic Encryption and Privacy-Preserving Machine Learning**. *M.S. Thesis* 2023. [arXiv:2310.13140](https://arxiv.org/abs/2310.13140)

TEACHING & INVITED TALKS

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| Understanding the Failure Modes of Transformers through the Lens of GNNs | Fall 2026 |
| <i>Invited Lecture · SMU Lyle School of Engineering & Guildhall</i> | |
| <ul style="list-style-type: none">90-minute lecture on information-propagation issues in causal transformers, viewed through the graph neural network (GNN) theory of oversquashing and oversmoothingGentle introduction to GNN theory and its connection to causal transformers (causal masking as a directed, bipartite graph neural network) | |
| written version | |
| Improving the Attention Mechanism with Dynamic Relational Priming | Spring 2025 |
| <i>Invited Lecture · SMU Lyle School of Engineering & Guildhall</i> | |
| <ul style="list-style-type: none">90-minute lecture on the ICML 2026 paper "Dynamic Relational Priming Improves Transformer in Multivariate Time Series"Introduction to multivariate time series, its challenges in deep learning, and existing solutions | |
| Analysis of Different Attention Mechanisms | Spring 2025 |
| <i>Invited Lecture · SMU Lyle School of Engineering & Guildhall</i> | |
| <ul style="list-style-type: none">90-minute lecture on transformer variants, from linear and sparse attention to emerging variants such as differential and selective attention | |
| written version | |

Introduction to Graph Neural Networks

Fall 2024

Invited Lecture · SMU Lyle School of Engineering & Guildhall

- 90-minute lecture on graph machine learning — leveraging graph structure in data (molecules, social networks, traffic) for structured learning
- Covered mathematical frameworks and a coding demo of widely used graph ML models
[code demos](#)

Privacy-Preserving Machine Learning in Neural Networks and Decision Trees

2023

Invited Lecture · CS3341 Foundations of Modern Computing, SMU Lyle School of Engineering

- Talk based on "Blind Evaluation Framework for Fully Homomorphic Encryption and Privacy-Preserving Machine Learning"
- Blind evaluation paradigm for encrypted programming

PH.D. COURSEWORK

GPA 3.8 / 4.0 (*A is the highest grade awarded at SMU*)

Course	Instructor	Grade
Machine Learning and Neural Networks	Eric C. Larson	A
Artificial Intelligence	Michael Hahsler	A
Data Mining	Miju Ahn	A
File Organization and Database Management	King Ip Lin	A
Advanced Applications Programming	Corey Clark	A
Computer Architecture	Theodore Manikas	A
Operating Systems and System Software	King Ip Lin	A